

quickstart

Quickstart & Validation Guide — Knowledge Areas & Bidirectional Deep Linking

Feature: specs/002-knowledge-areas-deep-linking | **Date:** 2026-09-01

How to verify this feature is actually done, rather than looking done. Every success criterion appears below with the command or procedure that measures it. A criterion with no entry here is a specification defect — feature 001 shipped one and it was found late, which is why this document exists as a Phase 1 artifact rather than an afterthought.

0. Before anything: re-derive the starting state

None of the figures in the specification may be quoted from a document. Each is re-derivable, and a stale figure is worse than none because it will be believed.

```
# The platform is reachable and answering
curl -s -o /dev/null -w '%{http_code}\n' "$WORKSHOP_BASE/api/health"

# Corpus composition — 1,101 / 1,055 / 44 / 2, and 267 uncertain
  when last measured
jq -r .kind workshop/curriculum/passages.jsonl | sort | uniq -c
jq -r 'select(.uncertain) | .pid' workshop/curriculum/
  passages.jsonl | wc -l
wc -l < workshop/curriculum/passages.jsonl

# Word-timing availability, per chapter — NOT corpus-wide
  (contract V2)
jq '.counts' workshop/curriculum/chapter-*/transcript.words.json
```

```

# The extracted taxonomy – counts and coverage, re-derived not
  quoted
jq '.track_count, .module_count' workshop/curriculum/chapter-*/
  knowledge/areas.json
jq '.term_count' workshop/curriculum/chapter-*/
  knowledge/terms.json
jq '.transcript_coverage' workshop/curriculum/chapter-*/
  knowledge/coverage.json
wc -l < workshop/curriculum/chapter-*/knowledge/linkage.jsonl #
  must equal the passage count

# The endpoints this feature adds do not exist yet
for p in /api/areas /api/terms /api/areas/x/questions; do
  printf '%s -> %s\n' "$p" "$(curl -s -o /dev/null -w '%
    {http_code}' "$WORKSHOP_BASE$p")"
done

# Export and diagram tooling – probe CAPABILITY, never just the
  name (contract V1)
for t in pandoc weasyprint pdftotext mmdc tesseract; do
  printf '%-12s %s\n' "$t" "$(command -v "$t" || echo ABSENT)"
done

```

The last one is not a formality. Every one of those tools resolves into a *user-local* directory on this host, and this repository has already shipped a defect where a name on PATH was a symlink into an unrelated cache: the binary answered a version query happily and rejected the flag that mattered. A probe that tests for a name reports a tool that is not there.

1. Validating the taxonomy (US1)

Criterion	How to measure it
SC-001 — every published area and term has evidence	Join the taxonomy to the mention set; count entries with zero mentions. Must be 0 . Then run G-KG-10 : publish one area with its mentions removed and confirm the check goes red.
SC-002 — extracted terms are genuine, not noise	Draw ≥ 30 terms with a recorded seed , have a reviewer judge each, require $\geq 90\%$. Store the seed with the judgements so the sample is reproducible and the verdict is challengeable. A sample drawn after seeing the result is not a measurement.

Criterion	How to measure it
SC-003 — every passage is accounted for	Assert <code>attached + classified_unattached == registry_count</code> , exactly. Then publish the attached proportion . Do not compare it to a threshold — none exists, and research U2 says why.
SC-004 — extraction is idempotent	Run twice over unchanged content; diff the taxonomy file byte-for-byte; assert 0 identifiers reassigned. Then G-KG-13 : make one identifier depend on content and confirm red.

The judgement call to watch for: SC-003 is the criterion most likely to be quietly “improved” by lowering what counts as attached. The check is arithmetic — a total, not a ratio — precisely so that it cannot be satisfied by redefining the numerator.

2. Validating the materials (US1)

Criterion	How to measure it
SC-005 — every area has all seven skeleton sections	Heading-presence check across every area source document. Any missing section fails.
SC-006 — every substantive claim is cited or marked editorial	Parse each document into claims, resolve every citation, assert no claim is both unmarked and uncited.
SC-007 — nothing is published without a review	Enumerate published areas; assert each has a review newer than the materials it reviews . A stale review fails; it does not warn.

3. Validating the deep links (US2) — the heart of the feature

Criterion	How to measure it
SC-008 — every relationship is bidirectional	Enumerate every relationship, traverse it in reverse through the platform's own interface, count failures. Must be 0 . Not a sample — the whole set, because a one-way link is indistinguishable from a two-way one when read forward.
SC-009 — media links land inside the cited span	For every media-backed citation assert $t_start \leq landed < t_end$. Publish the precision split — how many resolved at word precision and how many at segment — alongside the pass rate. A 100% pass rate at segment precision and at word precision are different products.
SC-010 — links survive correction and insertion	Replay the identifier-survival experiment already measured for passages, extended to the five new identifier kinds. Compare the full link set before and after by hash .
SC-011 — unresolvable links fail loudly	Delete a target; assert a loud outcome. G-KG-8 paired mutation: make the resolver re-point by nearest match; confirm red.
SC-012 — redaction reaches everything	Redact a passage; assert its text is absent from all eight propagation targets. G-KG-7 paired mutation: skip exactly one target; confirm red.

How SC-009 goes wrong silently: a test that only asserts “landed inside the span” passes equally for a word-precise link and a link that jumped to the start of a twenty-second segment. The precision split is what distinguishes them, which is why it is published rather than summarised.

4. Validating search (US3)

Criterion	How to measure it
SC-013 — every kind is actually searchable	For each kind, plant a known target and require a query to return it. Not a row count — a row count cannot distinguish a populated index from a populated table nobody queries, and the corpus already advertises one kind with zero entries.
SC-014 — every hit carries a resolving locus	Assert the property on every hit of the benchmark run. A hit without a resolving locus fails the run, not just that hit.
SC-015 — retrieval quality over the new kinds	A stored benchmark of ≥ 20 query-to-expected-identifier pairs; the gate prints per-query outcomes, not only the aggregate — an aggregate hides which queries fail, and which ones fail is the useful information.
SC-016 / SC-017 — latency survives corpus growth	Re-run the existing latency harness and publish the before and after together . A single after-figure cannot show a regression that stayed inside the threshold.

Also verify, though it is a contract rule rather than a criterion: the suggestion path still holds no embedder. Type-ahead must remain lexical; the structural guarantee is easy to lose by adding “just one” semantic lookup for the new kinds.

5. Validating assessment (US4) — where the workshop exceeds its reference

Criterion	How to measure it
SC-018 — every served question has a resolving citation	Enumerate every served question, resolve every citation, count failures. Must be 0 . Record the reference module's measured value for the same property — 0 of 785 — beside it, so the comparison is measured rather than claimed.
SC-019 — question-to-moment round trip	Traverse from question to cited moment and back for every eligible question; if exhaustive traversal is impractical, sample ≥ 30 with a recorded seed and publish the population size .
SC-020 — assembled answers are marked	Schema assertion over the whole set. An unmarked assembled answer fails.
SC-017a — short and long differ by depth	Assert both sets exist per area; assert every long-set question cites more than one distinct passage.
SC-017b — every lesson section is assessed	Join sections to questions; count sections with zero. Must be 0 .
SC-017c — coverage is honest	Emit the per-area table. G-KG-12 paired mutation: replace it with a mean; confirm red.

6. Validating the surface (US5)

Criterion	How to measure it
SC-021 — accessibility and keyboard operability	Automated audit per view, zero Level A/AA violations; plus a keyboard-only traversal that visits every deep-link affordance and asserts each can

Criterion	How to measure it
	be reached, activated and returned from.
SC-022 — styling resolves through tokens	Run the design toolkit's own token conformance check and a literal-value scan. Both are required: the first proves the tokens are valid, the second proves they are actually used. Either alone passes a stylesheet that defines perfect tokens and ignores them.
SC-023 — nothing existing was lost	Enumerate the four existing capabilities before the change; assert each is still reachable after . A before-and-after list, written down first — not a recollection.
SC-029b — interaction decisions are sourced	Traceability check over the interaction inventory; an entry with no named research finding fails.

7. Validating export (US6)

Criterion	How to measure it
SC-024 — four formats, complete	File-existence matrix over every published area. The reference achieves 25 of 34 ; this criterion requires complete, and the comparison is stated so nobody later matches the reference and calls it done.
SC-025 — exports are reproducible	Re-export unchanged material; extract text and diff . Extraction rather than byte comparison, because embedded timestamps make byte equality unachievable for some formats — and a

Criterion	How to measure it
	criterion nobody can meet is worse than none.

8. Validating the pipeline (US7) — the criterion that proves repeatability

This is the one that distinguishes a finished chapter from a finished feature.

Criterion	How to measure it
SC-015a — all six traversals exercised	For each row of the connectivity matrix, pick every eligible origin and assert a result, or an explicit “none exists”. A row with zero exercised origins fails — an unexercised traversal is unmeasured, not passing.
SC-015b — nothing is keyed on time	Inspect every stored relationship for its join key. G-KG-6 paired mutation: introduce one time-keyed link; confirm red.
SC-015c — a new chapter needs no hand-assembly	Run the documented procedure against a small synthetic chapter ; assert every output exists; assert the diff contains no manually created structural file and no code change .
SC-015d — the taxonomy is updated, not rebuilt	Capture the taxonomy before and after; assert 0 established identifiers changed; assert the contradiction report is non-empty when a contradiction was seeded.
SC-015e — incomplete never ships as complete	Withhold one required input; assert the run names exactly what is missing and publishes nothing.

Use a **synthetic chapter, not the real one**. Re-running the real chapter tests idempotency, which is SC-004. It cannot test whether a *new* chapter works, because every identifier it needs already exists — the run would pass by matching, never by minting, and the minting path is exactly what a second chapter exercises.

9. Validating honesty and governance

Criterion	How to measure it
SC-026 — limits document names every open defect	Cross-check recorded defects against the limits document; an unnamed defect fails. Paired mutation : remove one; confirm red. The two that must appear are the measured fabrication rate on unanswerable questions and the undefended topically-related-but-non-answering case.
SC-027 — every check has a paired proof	The check registry. Note it prints known debt on every run by design — a zero exit means every check is accounted for, not that every check has a paired proof. Read the debt rows.
SC-028 — every check is three-valued	Drive each into a could-not-determine condition — remove a dependency, make a service unreachable — and assert the third state, not either of the other two.
SC-029 — no workshop content in public repositories	The content-boundary check across the public surface, including this specification directory.
SC-029a — no reference content in the workshop	The same check run in the opposite direction . G-KG-16 paired mutation: scope it to outbound only; confirm red.
SC-030 — no server-side CI anywhere	The existing fleet-wide gate.

10. The eighteen gates and their proofs

Every gate in `contracts/knowledge-graph.md` §8 owes a paired mutation that turns it red. They are **owed**, not assumed — listed here so that a missing one is visible rather than discovered.

G-KG-1 route manifest completeness entity never served	G-KG-10 unevidenced
G-KG-2 unprovenanced question withheld kind actually retrievable	G-KG-11 advertised
G-KG-3 long-set citation breadth area, never a mean	G-KG-12 coverage per
G-KG-4 precision declared zero mints on re-run	G-KG-13 idempotency:
G-KG-5 offsets populated or field absent reported, not merged	G-KG-14 contradiction
G-KG-6 no time-keyed relationship fraction measured	G-KG-15 unjoined
G-KG-7 redaction reaches all eight targets boundary, BOTH directions	G-KG-16 content
G-KG-8 unresolvable link fails loudly chapter, no hand-assembly	G-KG-17 synthetic
G-KG-9 export unavailable $\Rightarrow$ state 2 names what is missing	G-KG-18 incomplete

Every proof must include a case that runs the real entry point against the real tree. This repository has twice shipped a proof that could not fail: one whose control step failed so zero mutations were ever reached, and one that exercised only sandboxed copies while the real entry point could not start. Both reported green.

11. What this feature will not be able to claim

Stated here so it is not discovered during a review.

- **Answering is not fixed by this feature.** The measured fabrication rate on unanswerable questions stands, and the topically-related-but-non-answering case remains undefended unless the third clarification is resolved towards closing it. Retrieval over the new kinds is safe because it refuses rather than generates; generation about the new kinds is gated.
- **Word precision is available, not uniform.** 17.4% of words fall below the recorded confidence threshold and 0.48% of words join to no segment at all. Every mention states which precision it used; none claims more.

- **No coverage threshold is asserted.** SC-003 and SC-017c publish measured figures with no target. Extraction has since measured chapter 01 at **83.79% of transcript passages** and **91.54% of words** covered — and that is a *measurement*, not a bar the next chapter must clear. The moment a measured figure becomes a target it starts being optimised toward, which is why the criterion requires publication rather than a threshold.
- **No area count is promised.** Extraction has since produced **7 tracks, 37 modules and 137 terms** for chapter 01 (research U3, settled). That 37 sits near the reference's 34 is a coincidence of magnitude and **must not be cited as parity** — one is an interview-preparation corpus, the other an extraction from a single working session, and they are not comparable quantities. No count was aimed at, and matching the reference's would have meant inventing coverage.
- **Prose generation cannot run on this host.** There is no generative model here. Two of the three authorship options work today; the third needs an operator action first.